

Question

The basic process of crude salt purification is as follows:

Dissolve 10 g of crude salt in 40 mL of deionized water with heating, then add 25% BaCl₂ solution to it until no more precipitate is produced, keep warm for 10 min, and perform the first normal pressure filtration. Then add saturated Na₂CO₃ solution until no more precipitate is produced, and perform the second normal pressure filtration. Add 6 M hydrochloric acid, then evaporate by heating in a water bath, filter under suction to remove the mother liquor, and finally dry by stir-frying to obtain refined salt.

Assume that 10 g of crude salt is composed of 9 g NaCl, 0.3 g MgCl₂, 0.3 g CaCl₂, 0.2 g FeCl₃ and 0.2 g Na₂SO₄. It is known that BaCl₂ and Na₂CO₃ are added with a 50% excess, 1 mL of solution is lost in each normal pressure filtration, and 5 mL of mother liquor is lost in vacuum filtration. Estimate the theoretical yield and theoretical yield rate based on the following solubility data.

Hint: It is considered that the volume of the solution remains unchanged and remains at 40 mL before evaporation and concentration, that is, deionized water will be added after the volume decreases (such as after normal pressure filtration), and the excess water will evaporate after the volume increases; vacuum filtration is performed immediately after evaporation and concentration, and it can be considered that the temperature at this time is about the boiling point of water; it is considered that 1 mL of aqueous solution contains 1 mL of water.

The data in the table below is the mass fraction of solute in saturated NaCl solution.

Temperature	Mass Fraction
25°C	26.45%
100°C	28.05%

A. All other options are incorrect

B. 5.69 g, 56.9%

C. 6.24 g, 62.4%

D. 7.70 g, 77.0%

E. 8.16 g, 81.6%

F. 8.54 g, 85.4%

G. 9.11 g, 91.1%

H. 5.94 g, 66.0%

I. 6.73 g, 74.8%

J. 7.32 g, 81.3%

K. 7.92 g, 88.0%

L. 8.34 g, 92.7%

M. 8.70 g, 96.7%

N. 9.33 g, 103.7%

Answer

Correct Answer: **E**

Detailed Explanation

The composition of the initial solution is shown in the following table:

Na^+	Cl^-	Mg^{2+}	Ca^{2+}	Fe^{3+}	SO_4^{2-}	Ba^{2+}	
n/mmol	156.8	169.4	3.151	2.703	1.233	1.408	0

After adding an excess of 50% BaCl_2 , SO_4^{2-} is converted to BaSO_4 .

Na^+	Cl^-	Mg^{2+}	Ca^{2+}	Fe^{3+}	SO_4^{2-}	Ba^{2+}	
n/mmol	156.8	173.6	3.151	2.703	1.233	0	0.704

CHECKPOINT

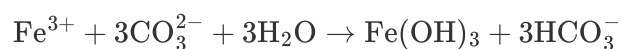
1 PTS

The amounts of substance of Cl^- and Ba^{2+} after adding an excess of 50% BaCl_2 are 173.6 mmol and 0.704 mmol, respectively.

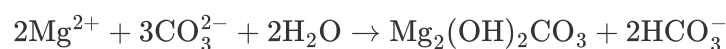
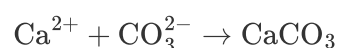
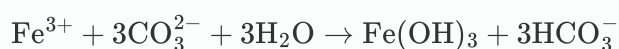
After filtration with a loss of 1/40, the composition becomes:

Na ⁺	Cl ⁻	Mg ²⁺	Ca ²⁺	Fe ³⁺	SO ₄ ²⁻	Ba ²⁺	
n/mmol	152.9	169.3	3.072	2.635	1.202	0	0.686

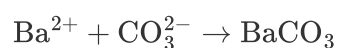
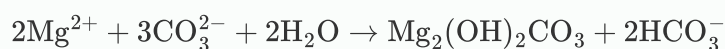
Then, an excess of 50% Na₂CO₃ was added. First, write out the equations for the reactions of each ion:


CHECKPOINT

0.5 PTS


CHECKPOINT

0.5 PTS



Therefore, the stoichiometric ratio is $1\text{Fe}^{3+} - \frac{3}{2}\text{CO}_3^{2-}$, $1\text{Ca}^{2+} - 1\text{CO}_3^{2-}$, $1\text{Mg}^{2+} - 1\text{CO}_3^{2-}$, $1\text{Ba}^{2+} - 1\text{CO}_3^{2-}$

At this point, only focus on Na⁺ in the reaction. After adding Na₂CO₃, all metal cations in the solution except Na⁺ are converted to the corresponding precipitates, but the total amount of charge carried by their anions remains unchanged. Therefore, the amount of substance of cations can also be determined by charge conservation. Then, according to the stoichiometric ratio, Na⁺ in the solution is:

$$152.9 + (2 \times 3.072 + 2 \times 2.635 + 3 \times 1.202 + 2 \times 0.686) \times 1.5 = 177.5 \text{ mmol}$$

CHECKPOINT

1 PTS

The amount of Na^+ in the solution after adding Na_2CO_3 is 177.5 mmol

Then, after filtration with a loss of $1/40$ and addition of concentrated hydrochloric acid, 173.1 mmol NaCl is obtained, which is 10.11 g NaCl .

CHECKPOINT

1 PTS

There is 10.11 g NaCl in the solution after adding concentrated hydrochloric acid

Then, vacuum filtration loses 5 mL of saturated NaCl solution, and the mass of NaCl in it is $5 \text{ g} \times \frac{28.05\%}{1-28.05\%} \approx 1.95 \text{ g}$.

CHECKPOINT

1 PTS

1.95 g NaCl is lost by vacuum filtration

Therefore, the final theoretical yield is 8.16 g , and the theoretical yield is 81.6%.